

Dear Editors,

The authors have described the proposed book to me and asked whether I would contribute. I am writing to support the proposal and confirm my intention to participate.

For the past 30 years, I have led product and technology development projects involving high-speed electrical machines in compressors for air, process gases and refrigerants. My scientific background is in their loss mechanisms, particularly aerodynamic and friction losses in the air gap. I now work at Upheat on high-temperature industrial heat pumps, where these machines are designed and sold rather than merely studied. This combination of research and industrial experience informs two observations.

The first concerns air-gap losses. Below a surface speed of approximately 100 metres per second, the fluid in the gap can often be treated as an inert medium and windage as a minor part of the loss budget. Above this level, however, the air gap becomes an active loss mechanism and, in some machines, the largest single source of loss. Only a few correlations are available for predicting these losses, each validated over a narrow range of Taylor numbers, clearances and surface conditions. In practice, they are frequently applied outside their validated ranges because no better methods exist. A book that presents these correlations together with the conditions under which they were developed—and states clearly where they cease to be reliable—would prevent considerable misplaced confidence.

The second observation concerns the gap between industrial practice and the design literature. Gearless high-speed drives became commonplace in turbomachinery before comprehensive design texts were available. The transition was driven by component availability, oil-free operation and the elimination of gearbox maintenance. Much of the engineering knowledge was developed within companies, one machine at a time. An engineer entering the field today will find many research papers, but no coherent design procedure. This book proposes to provide one and demonstrate it by following two machines from specification to measured performance.

Subject to the publishing contract, I intend to write the chapter on applications and system architectures. It would cover the transition to direct-drive technology; gearless compressors, blowers and organic Rankine cycle units; subsea and hermetic process machines; and high-temperature industrial heat pumps, which are currently a major driver of high-speed compressor development. It would also explain how architectures are selected by balancing total cost of ownership against specific power.

I would invite co-authors to contribute expertise on mobile and aerospace applications and aerodynamic optimisation, while taking responsibility for the coherence of the chapter. My long experience in high-speed-machine design and loss analysis would allow me to bring both technical depth and a current industrial perspective to the book.

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